

CHENGLONG YU

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EDUCATION

Institute of Computing Technology, Chinese Academy of Sciences (ICT, CAS) Beijing, China
Master of Engineering in Computer Technology Sep. 2024 – Expected Jun. 2027
China University of Petroleum (East China) Qingdao, China
Bachelor of Engineering in Computer Science and Technology Sep. 2020 – Jun. 2024
GPA: 4.12/5 | Rank: 3/114 (Top 3%)

RESEARCH INTEREST

World Models, Reinforcement Learning, Diffusion Models, Knowledge Distillation

RESEARCH EXPERIENCE

World Model-Guided Multi-Agent Reinforcement Learning (MARL)

Independent Research Dec. 2023 - Nov. 2025

- **Identified** the low sample efficiency and slow convergence in traditional MARL caused by heavy reliance on trial-and-error interactions with real environments.
- **Proposed** a World Model-Guided MARL that utilizes a world model to simulate environment dynamics and accelerate agents' policy exploration.
- **Applied** fine-grained constraints on the action space during the training phase, which reduced the policy search space and further accelerated algorithm convergence.
- **Achieved** a 36% improvement in convergence speed compared to baseline algorithms while yielding superior control policies.

Diffusion Model-Assisted Sample-Efficient Data-Free Knowledge Distillation (DFKD)

Independent Research Aug. 2025 - Present

- **Identified** a critical limitation in existing DFKD methods, which prioritize synthetic sample diversity but fail to ensure the student model acquires core semantic knowledge from these samples.
- **Proposed** an attention-constrained framework that aligns the student's focus with the teacher's, applying targeted data augmentation strictly to background areas to preserve core features. **Leveraged** GLIDE to augment the synthetic dataset, resolving the lack of semantic information in GAN-generated images.
- **Achieved** SOTA accuracy on the CIFAR-10 and CIFAR-100 using the proposed distillation approach.

PUBLICATIONS & PATENTS

- **C.Yu**, W. Xing, J. Shi, Y. Zhou and L. Liu, "Collaborative Multi-Agent Deep Reinforcement Learning for Joint Task Offloading and Resource Allocation with Long Term Energy Control," 2025 IEEE 102nd Vehicular Technology Conference (VTC2025-Fall), Chengdu, China, 2025, pp. 1-5.
- **C.Yu**, Y. Zhou, Y. Qi and J. Wang, "Joint Task Offloading and Resource Allocation with Multi-Task Energy Control based on World Model Guided Collaborative Multi-Agent Deep Reinforcement Learning," IEEE Internet of Things Journal (under review).
- Y. Zhou, **C. Yu**, J. Shi, Y. Qi, W. Xing, and L. Liu, "An Industrial Internet System for Resource Allocation Based on Collaborative Multi-Agents." Chinese National Invention Patent, Patent No.: CN202511002125.2.
- Y. Qi, Y. Wu, Y. Zhou, **C. Yu**, et al. "Agent-Oriented Multi-Mode Convergence and AI-Native Orchestration for Low-Altitude Intelligent Networks," IEEE Communications Magazine (accepted).

HONORS & AWARDS

- Outstanding Graduate of Shandong Province - 2024
- National Encouragement Scholarship (Three times) - 2021, 2022, 2023
- UCAS Academic Scholarship (Twice) - 2024, 2025
- Honors Bachelor's Degree, China University of Petroleum (East China) - 2024

TECHNICAL SKILLS

- **Technology stack:** Python, C, C++, Pytorch, Linux, Git, CUDA